

Alaf Nascimento

Paris, France

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EDUCATION

2024 – present - Doctor of Philosophy in Mathematics and Computer Science

Télécom Paris, Institut Polytechnique de Paris, Palaiseau, France.

- Thesis Title: Real-Time Supervisory Control and Data Acquisition Systems via 5G Standalone.
- Specialization: Computer Science, Data, and Artificial Intelligence (AI).
- Relevant Modules: *Advanced Topics in 5G-6G RAN for IIoT, Research Ethics, ORCID IDs.*

2023 – 2024 - Master's in Embedded Systems and Information Processing, *summa cum laude*

Télécom Paris, Institut Polytechnique de Paris, Palaiseau, France.

- Grade: 4.0 GPA (4.0 scale).
- Research project: Modeling the Critical Real-Time Execution of a 5G Base Station.
- Specialization: Industrial Systems and Embedded Artificial Intelligence.
- Relevant Modules: *Embedded Linux, Embedded Artificial Intelligence, Artificial Intelligence for Robotics, Language Processing, System Modeling, and Real-time Systems.*

2017 – 2024 - Engineering Dual Degree Program

- **2022 – 2024 - Master of Science in Engineering**

Télécom Paris, Institut Polytechnique de Paris, Palaiseau, France.

- Grade: 3.8 GPA (4.0 scale).
- Thesis Title: Remote Access Application for Matter IoT Devices.
- Specializations: Embedded Systems and Mobile Networks.
- Relevant Modules: *Reconfigurable Architectures (FPGA), Concurrent Programming, Microprocessor-Based Systems, IoT Protocols and Systems, and Mobile Networks.*

- **2017 – 2022 - Bachelor of Science in Electrical Engineering**

Federal University of Espírito Santo, Vitória, Brazil.

- Grade: 3.7 GPA (4.0 scale).
- Thesis Title: Multiplatform System For Data Reception Via Visible Light Communication.
- Specializations: Electronics, Telecommunications, Mechatronics, and Computer Science.
- Relevant Modules: *Embedded Systems, Computer Architecture, Digital Systems, Computer Networks, Mobile Robotics, Computer Vision, and Oriented IoT Project.*

PROFESSIONAL EXPERIENCE

2025 - Teaching Assistant

Télécom Paris, Institut Polytechnique de Paris, Palaiseau, France.

- Practicals on the Rust programming language.

2024 - Embedded Systems and IoT Intern

Orange S.A., Meylan, France.

- Remote access application for Matter IoT devices.
 - Implementation of a tool for transforming a Matter data model into the USP data model
 - Embedded software for an IoT gateway based on ARM Cortex-A processor.
 - Prototype showing the capabilities of a Matter device using the USP protocol
- Developed the first version of the USP data model capable of interacting with the Matter protocol. Achieving an A+ in my master's thesis.

2023 - Network and Automation Intern.

Synchrotron SOLEIL, Saint-Aubin, France.

- Software tool parameterization dedicated to centralized supervision of Siemens PLCs (S7-3xx and S7-15xx).

- Over 98 % of the targeted devices were covered through a solution based on the S7 and SNMP protocols.
 - Real-time monitoring tool: Zabbix;
 - Programming languages: Python, C/C++, and CMake.

2021 – 2022 - Embedded Systems and IoT R&D Intern

2Solve Engineering and Technology, Vitória, Brazil.

- Development of software for embedded systems, IoT Web Applications, and technical documentation.
 - Embedded systems based on Raspberry Pi and SAMD21.
 - Programming languages: Javascript, Python, C/C++, and CMake.
 - Dev tools: NodeJS, AngularJS, InfluxDB, and MongoDB.
 - IoT tools: Node-RED and Grafana.
- Research project: Design of an OOK transmitter for short-link visible light data communication.

2019 – 2021 - Undergraduate Student Researcher

UFES Telecommunications Laboratory (LabTel), Vitória, Brazil.

- Software and hardware design for visible light communication systems (VLC systems).
 - Dev tools: Android Studio, NodeJS, VueJS.
 - Programming languages: MatLab, Java, Python, and C++.
- Research projects:
 - Application of Visible Light Communication Technology in Monitoring High-Risk Newborns;
 - SmaL: Smartphone Receiver for Coded Data via Light.
 - Publications: [1], [2], and [3].

2019 – 2020 - Automation Developer

Cassiano Antonio Moraes University Hospital (HUCAM), Vitória, Brazil.

- Establishment of electronics for a supervisory system, data monitoring app, and creation of technical documentation.
 - Embedded systems based on Raspberry Pi, Arduino, and ESP8266.
 - Real-time monitoring tool: Zabbix.
 - Programming languages: Python, Javascript, and C++.
 - Publications: [4]

2019 - Educational Program Fellow

Tutorial Teaching Program (PET), Vitória, Brazil.

- Group of distinguished students from the Electrical Engineering department at UFES. Software training, such as LaTeX. Research about embedded systems. Production of scientific articles.
 - Embedded systems based on Raspberry Pi and Arduino;
 - Programming languages: MatLab, Python, and C/C++.
 - Publications: [5]

MENTORING EXPERIENCE

2025 - Undergraduate Final Project Jury Member

Federal University of Espírito Santo, Vitória, Brazil.

- Project: Video streaming in Kubernetes.
- Author: Gustavo Teixeira Acioli.

VOLUNTEERING

2018 – 2019 - Activity Manager

Academic Center of UFES Electrical Engineering, Vitória, Brazil.

- Organization of welcome events for freshmen, lectures on subjects of interest to graduation, promotion of sports events, selling of engineering custom t-shirts, and maintaining the study room.

2018 - Museum Mediator

UFES Museum of Life Sciences, Vitória, Brazil.

- Introduce the museum to visitors, control the flow of people, and pass safety guidelines.

HONOURS AND AWARDS

2025 - Honorable Mention, Find Me on the Moon: NASA Lunar Navigation Challenge

National Aeronautics and Space Administration (NASA), USA.

- Selene Squad: rover designed for navigating, mapping, and characterizing the Shackleton Crater.

2022 – 2024 - BRAFITEC scholarship

CAPES Foundation, Brazil.

- Engineer's degree funding granted based on the criteria of academic and technical excellence.

2016 - Honorable Mention, Brazilian Public School Mathematics Olympiad

Institute of Pure and Applied Mathematics (IMPA), Rio de Janeiro, Brazil.

- Stood out in mathematics at this Olympiad, being the only high school student out of around 500 in the school to receive this award.

2015 – 2016 - Outstanding certificate at the São João Batista School Science Fair

EEEFM São João Batista (High School), Espírito Santo, Brazil.

- 2016 (1st place) - Tesla coil capable of creating electric arcs of a few centimeters.
- 2015 (2nd place) - Tesla coil capable of wirelessly turning on fluorescent lamps.

LANGUAGE SKILLS

- Portuguese - Native.
- English - Advanced (C1, 2024).
- French - Advanced (C1, 2024).
- Spanish - Intermediate (B1, 2025).

PUBLICATIONS AND APPEARANCES

1. **NASCIMENTO, A., 2022. Multiplatform System For Data Reception Via Visible Light Communication.** Bachelor's thesis. Federal University of Espirito Santo, ES/Brazil.
2. ZWAAG, K., ROCHA, H., SEGATTO, M., BASTOS, T., SILVA, J., SANTOS, F., **NASCIMENTO, A.** et al., 2021. **Performance Evaluation of an OOK-Based Visible Light Communication System for Transmission of Patient Monitoring Data.** IFMBE Proceedings.
3. **NASCIMENTO, A.,** ROCHA, H., SEGATTO, M., BASTOS, T., SILVA, J., ZWAAG, K. et al., 2020. **Application of Visible Light Communication Technology for Monitoring in Hospitals.** Brazilian Congress on Biomedical Engineering.
4. **NASCIMENTO, A.,** JUNIOR, L., JARDIM, I., 2020. **Low-Cost Module for Supervisory System of Hospital Substations.** In: Congresso Internacional Online das Engenharias.
5. JURESWKI, A., **NASCIMENTO, A.,** MENDONÇA, M., ULHOA, P., 2020. **History of PET Electrical Engineering UFES.** Brazilian Congress of Engineering Education.